est (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

ANNUAL REVIEWS

Annual Review of Animal Biosciences

The Rhesus Macaque as an Animal Model for Human Nutrition: An Ecological-Evolutionary Perspective

Zhenwei Cui,[1,2] Yunlong Dong,[1,2] Jonathan Sholl,[3] Jiqi Lu,[2] and David Raubenheimer[1,4,*]

[1]Centre for Nutritional Ecology, Centre for Sport Nutrition and Health, School of Physical Education (Main Campus), Zhengzhou University, Zhengzhou, Henan, China

[2]Institute of Biodiversity and Ecology, Zhengzhou University, Zhengzhou, Henan, China

[3]ImmunoConcept Lab, Université de Bordeaux, Collège Sciences de la Santé, CNRS UMR 5164, Bordeaux, France

[4]Charles Perkins Centre, School of Life and Environmental Sciences, University of Sydney, Sydney, New South Wales, Australia; email: david.raubenheimer@sydney.edu.au

ANNUAL REVIEWS CONNECT

www.annualreviews.org

- Download figures
- Navigate cited references
- Keyword search
- Explore related articles
- Share via email or social media

Annu. Rev. Anim. Biosci. 2025. 13:441–64

First published as a Review in Advance on November 18, 2024

The *Annual Review of Animal Biosciences* is online at animal.annualreviews.org

https://doi.org/10.1146/annurev-animal-111523-102354

Copyright © 2025 by the author(s). This work is licensed under a Creative Commons Attribution 4.0 International License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited. See credit lines of images or other third-party material in this article for license information.

*Corresponding author

OPEN ACCESS

Keywords

nutritional ecology, animal model systems, obesity research, caloric restriction, nutritional geometry, niche models

Abstract

Nutrition is a complex and contested area in biomedicine, which requires diverse evidence sources. Nonhuman primate models are considered an important biomedical research tool because of their biological similarities to humans, but they are typically used with little explicit consideration of their ecology and evolution. Using the rhesus macaque (RM), we consider the potential of nutritional ecology for enriching the use of primates as models for human nutrition. We introduce some relevant aspects of RM evolutionary and social ecology and discuss two examples where they have been used in biomedical research: obesity and aging. We next consider how insights from nutritional ecology can help inform and direct the use of RM as a biomedical model. We conclude by illustrating how conceptual tools might inform the use of RM as a model for human nutrition and extracting insights from RM that might be relevant to broader theoretical considerations around animal model systems.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

INTRODUCTION

Among the most important, complex, and contested areas in biomedical and public health research is nutrition. Malnutrition (overnutrition, undernutrition, and imbalanced nutrition) remains a leading public health concern, with poor diets being responsible for one in five preventable deaths worldwide (1). Despite substantial investment and research effort, there is little evidence that the situation is improving. This has led to a proliferation of competing theories and critical attention to the nature, scope, and appropriateness of methods applied in nutritional research (2–5).

Model organisms are one of the most widely used tools in the biological and biomedical sciences, largely for their potential to provide a simplified representation of biological systems that for ethical, practical, or scientific reasons cannot be studied directly on humans or other target species (6, 7). Taxonomically, their reach spans the diversity of life, from viruses, yeast, and bacteria to plants, invertebrates, rodents, and primates, potentially including humans (8). The contributions of model organisms to basic biology have been immense, with much of what we know about cell biology, endocrinology, inheritance, neurophysiology, development, and many other fundamental biological processes attributed to their study, as well as important methodological advances, such as the tools for sequencing the human genome. Added to the indirect benefits of such advances, model organisms have made many important direct contributions to nutrition science and are considered essential for unraveling the complexities of organism–food interactions (9, 10).

The use of model organisms is, however, not without its complexities and controversies. Indeed, the ubiquitous use of the modifier "model" before terms such as organism, species, animal, and system has become one of the "most overused and underdefined words in all of life sciences" (11, p. 649). So, what is a model organism/system? There is no single answer to this question—its definition varies with the field and context of research, species involved, goals of the study, and other factors (12). To some working biologists, especially those whose research involves established model organisms such as nematode worms, fruit flies, laboratory rodents, or rhesus macaques (*Macaca mulatta*), such considerations might seem arcane, even diversionary, given the proven track records of their study subjects. However, ethical, theoretical, and practical considerations increasingly raise questions of whether model animals are an optimal, or even necessary, tool to address a research question, and if so, which model is best suited for the purpose (13–16). Nowhere are these considerations more important than in the use of nonhuman primate models (17), of which rhesus macaques (henceforth RM) are among the most widely used in biomedical research (18, 19).

In this review, we integrate research across disciplinary boundaries and consider the potential role of nutritional ecology for contributing to the use of RM as an animal model in human nutrition. We begin by introducing some relevant aspects of RM evolutionary and social ecology and thereafter discuss examples where RM have been used in biomedically focused nutritional research, with a focus on obesity and aging. We then develop insights from nutritional ecology that can inform and direct nutrition-related biomedical research using RM. We conclude with a brief discussion of conceptual tools that might inform the use of RM as a model for human nutrition and reciprocally extract insights from RM that might be relevant to broader theoretical considerations around animal model systems.

RHESUS MACAQUE EVOLUTIONARY AND SOCIAL ECOLOGY

RM are widely considered to share biological, ecological, and evolutionary traits with humans that make them particularly well suited as a model system for biomedicine (20, 21). In this section, we discuss some key aspects of the evolutionary and social ecology of RM that are potentially relevant to their use as a model organism for human nutrition. Central to this is the relatively close phylogenetic relation of RM to humans and the pronounced ecological, including dietary, flexibility

the two species show compared with most other primates that occupy relatively specialized niches (19).

The genus *Macaca* originated 7 million years ago, probably in North Africa, having diverged from a common ancestor with humans and chimpanzees approximately 23 million years ago (22), and over the past 5 million years underwent a burst of speciation in Asia (23). The 24 extant macaque species are classified into four species groups: the silenus, fascicularis, sinica, and sylvanus groups (23–25). The macaques are the most geographically and ecologically widespread of all nonhuman primates.

Cross-cutting the phylogenetic classification of macaques is an informal ecological classification into "weed" and nonweed species, based broadly on the dietary niche breadth and adaptability of different species (26). This ecological classification was proposed as a framework to explain the evolutionary spread, current distribution, and capacity of some macaque species to thrive in the ecological disturbance associated with humans, while emphasizing that it is not intended in the normative sense of a pest species requiring elimination (26). Most relevant here are the weed species, defined as the group that are well adapted to anthropogenic environments and closely associated with humans through much of their range: *Macaca fascicularis* (the long-tailed, also known as crab-eating, macaque), *Macaca radiata* (the bonnet macaque), *Macaca sinica* (the toque macaque), and *M. mulatta* (the RM). Like macaques, early hominids such as *Australopithecus* might owe their ecological and evolutionary success to being weed species (27).

Of all macaques, the RM are the most geographically widespread and ecologically diverse. Indeed, the native range of RM alone exceeds all other primates except humans, comprising more than 6,000 km^2 across 11 countries from India in the west to the east coast of China and south into Thailand and Vietnam (17). A typical weed species, RM readily adapt to disturbed and modified environments, such as secondary forests, agricultural landscapes, and (peri)urban environments (19). They are intimately associated with humans across their distribution, including their native range in Asia and their establishment across the globe in research colonies and in naturalized populations on other continents (28, 29). This broad distribution might be due partly to their ability to thrive within the rapidly expanded ecological disturbance created by humans (26, 30). However, evidence suggests they are not merely "disturbance specialists" but possess biological, behavioral, and social attributes that make them, in common with humans, more generally adaptable to diverse ecologies, natural and anthropogenically altered alike. First, RM are estimated to have dispersed across Asia between 150,000 and 200,000 years ago (18), prior to the widespread occurrence of humans. Second, their natural habitat spans elevations from sea level to 4,000 m, encompassing tropical, subtropical, and temperate environments, including evergreen deciduous forests, semideserts, swamps, grasslands, and mangroves, in contrast with most primate species that are ecologically more specialized. Third, like humans (31), RM are considered broad dietary generalist omnivores with a notably flexible and varied diet (19). Most populations are predominately herbivorous, but some have been documented to eat diets in which 15% of intake (20.6% of natural foods eaten) is composed of animals (predominately invertebrates) (32). Human societies, too, vary widely in the plant/animal balance of the diet, but the proportion of animal foods tends to be higher than documented for RM (33). Important for considerations of RM as model organisms for human nutrition is the question of which biological attributes underpin their dietary and general ecological adaptability and how these align with humans.

Paralleling their ecological diversity, RM have a high degree of morphological and genetic variation (34), with nucleotide diversity estimated from research colonies to be more than twice that of humans (18, 35). Based on morphological and genetic criteria, various studies have subdivided the species into between 6 and 13 subspecies (36–38). Most research into the origins and functional implications of this diversity, however, has focused on comparisons of Chinese- and Indian-origin

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

macaques, the two groups from which almost all research colonies originated. These populations, which are estimated to have diverged between 200,000 and ~150,000 years ago and have remained in relative isolation since (18, 39), are as genetically distinct as some primate species (40). A large amount of research has examined functional genetic variation between these populations from a biomedical perspective (19, 41), and more recently evolutionary aspects have come into closer focus (42).

An interesting population genomics study of wild RM encompassing much of the species range in China found that the populations are structured into genetic lineages corresponding with 5 recognized subspecies (41), estimated to have diverged between 125.8 and 51.3 thousand years ago. The study found evidence of functional divergence through the positive selection of genes associated with conservation and acquisition of energy. Consistent with the prediction of Bergmann's rule, there were signals of positive selection of genes associated with larger body size at higher latitudes, assumed to reduce heat loss through decreasing body surface area relative to volume. A positive association between latitude and body size in RM was documented previously (37), and one study reported evidence of a recent decrease in body size in a free-ranging research population established on the tropical island of Cayo Santiago in Puerto Rico, which was founded in 1938 from individuals translocated from a cooler climate in northern India (43). In addition to body size, Liu et al. (41) found evidence in the northernmost subspecies inhabiting the highly temperate Taihangshan mountain range for positive selection of genes associated with gluconeogenesis, the biochemical process in which glucose is produced from noncarbohydrate sources such as dietary amino acids. This is possibly an adaptation to surviving extreme cold and food shortages in winter, as we discuss below (see the section titled Insights from Nutritional Ecology).

Evidence thus suggests that a degree of genetic divergence among populations has played a role in RM's adaptation to diverse environments, as is the case for humans (44). Evidence also indicates that their social structure and behavioral flexibility contribute to their generalized ecology. RM are considered to have notably complex and flexible behavioral repertoires (45), and comparative studies have shown that some dimensions of social structure are particularly pronounced in RM relative to other macaques, and indeed other primate species, or shared with only one or a few other macaques. For example, RM societies are strongly nepotistic with steep dominance hierarchies, categorized as Grade 1 (least socially tolerant) among four macaque species groups that vary along the spectrum from despotic to egalitarian (45–47). Alongside RM in Grade 1 is the Japanese macaque (*Macaca fuscata*), which is also notable for its ecological generalism (48), including a broad geographic distribution within Japan, and cultural adaptability, evidenced through interactions with humans, including bathing in hot springs and potato washing (49). Nepotism has, likewise, been implicated in the adaptability of human societies (50).

Another social factor that has been implicated as important in the ecological adaptability of humans is nonkin food sharing (51). Although it might intuitively seem at odds with the despotic nature of RM social structure, nonkin food sharing has been documented in RM (52) and in another weed species, long-tailed macaques (53). In a comparison of four macaque species, Amici et al. (54) found that, contrary to expectations, social tolerance over food was higher in more despotic species, and in all four species studied, higher ranking and more socially integrated individuals achieved greater access to food. An interesting theory through which food sharing might link population adaptability to despotic social structure is that in steeply hierarchical societies, less dominant individuals with poor access to resources might have a greater need to experiment with novel foods (54).

Patterns of covariation between suites of social traits related to dominance and social tolerance, such as the high degree of nepotism and pronounced social hierarchy shared by RM and Japanese macaques, have been described as "social styles." Originally attributed to species'

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

phylogenetic history, social styles have more recently been reconceptualized as "social reaction norms," highlighting the complex interplay of genetic history and behavioral flexibility in shaping these patterns (55). This conceptual shift is important when considering the role of social systems in the capacity of species such as weed macaques to adapt to diverse environments, because it characterizes social systems themselves as a plastic trait that can potentially adapt over short timescales to environmental change. For example, Testard et al. (56) documented increases in social connectivity among free-ranging RM in the Cayo Santiago research colony following the devastating impact of Hurricane Maria in 2017. They concluded that their study supports the hypothesis that social cohesion is an important mechanism through which primates can adapt to even extreme environmental change. A recent follow-up study showed that increased social cohesion is adaptive in the disturbed ecology, with the degree of social tolerance predicting survival of individuals in the years following the hurricane but not before it (57).

This brief overview suggests that genetic, behavioral, and social factors might interact in enabling the ecological and dietary flexibility of RM, as is the case for humans (58). More research is needed to understand how these factors align in the two species and their relevance for the widespread use of RM as a model for biomedical research.

THE RHESUS MACAQUE IN NUTRITION-RELATED BIOMEDICAL RESEARCH

RM have been used as a model organism for biomedical research in a vast number of contexts, including many aspects of aging, drug and vaccine development, COVID, cancer, susceptibility to HIV infection and development of AIDS, transplantation research, reproductive endocrinology, stress responses, atherosclerosis, obesity, cancer immunotherapy, and many more. Although this prodigious range of biomedical uses, as well as the many aspects of basic biology that have been studied using RM, is important in the context of this review (see the section titled The Rhesus Macaque as a Model System for Human Nutrition below), in this section we focus specifically on two research areas that illustrate how ecological and evolutionary considerations can complement the use of RM as laboratory animals for nutrition-related biomedical research: obesity and aging. For broader discussions on basic and biomedical uses of RM as a model animal, we refer the reader to Phillips et al. (17) and Xue et al. (18).

Obesity and Associated Diseases

Nonhuman primates are considered important models for obesity research because their metabolic physiology more closely matches that of humans than do other model systems such as rodents (17, 59), with RM being the most widely used primate model for this purpose (60). Owing to the broad scope of this review, we are restricted to illustrative coverage, emphasizing issues most directly relevant to our ecological discussions below. For more detail, readers are referred to work by Havel et al. (59), Kleinert et al. (60), Wood & Sullivan (61), Kemnitz (62), Bauer et al. (63), and Mattison & Vaughan (64).

Like humans, RM are prone to develop spontaneous obesity associated with age and diet (65), which is in turn associated with metabolic problems such as insulin resistance, type 2 diabetes, dyslipidemia, and vascular dysfunction (66). Most research has concerned captive laboratory monkeys, of which approximately 10–15% become spontaneously obese when maintained on an ad libitum diet of standard laboratory biscuits (67). Spontaneous obesity and diabetes have also been detected in free-ranging macaques provisioned with a high-protein (26% of energy) commercial diet in the Cayo Santiago research colony (68), demonstrating that obesity in this species (and other research primates) is not simply due to reduced activity in captive conditions but is related to diet (33).

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

Several models have been proposed to explain the dietary causes of obesity in humans, including models that implicate the intake of energy per se (the energy balance model, EBM) (69), glucose (the carbohydrate insulin model, CIM) (70), fructose (the fructose survival hypothesis, FSH) (71), fat (the hedonic overdrive model, HOM) (72), and dietary protein dilution (the protein leverage hypothesis, PLH) (5, 73). The EBM proposes that obesity results when environmental circumstances (e.g., increased availability of hyperpalatable foods) interact with brain mechanisms and cause energy intake to increase to levels that exceed energy expenditure (positive energy balance), regardless of the nutritional source of that energy. The CIM, in contrast, proposes that obesity is driven specifically when the ingestion of high–glycemic load carbohydrates, such as glucose and sucrose, triggers fat storage, which in turn causes increased hunger and energy intake. Under the FSH, the principal nutritional driver of obesity is fructose, through reducing intracellular ATP levels, causing increased hunger and impaired satiety. According to the HOM, obesity results when high-fat diets stimulate the hedonic system in the brain, which overrides normal homeostatic systems, resulting in excessive intake and storage of energy. Finally, the PLH posits that the tight regulation of protein intake by humans results in incidental overconsumption of fats and carbohydrates on diets with a low protein:energy ratio. Although no biomedical research on RM of which we are aware has been performed specifically to discriminate among or reconcile these hypotheses, several studies are potentially relevant.

Similar to humans, captive RM developed central obesity, dyslipidemia, inflammation, and glucose intolerance (including, in some, overt diabetes) within a year of feeding on a high-fructose diet (74). This ostensibly aligns with the predictions of the FSH, EBM, and PLH, all of which predict increased adiposity on high-fructose diets, but not readily with the CIM, because fructose has a relatively low glycemic index. One study compared in captive middle-aged male RM the effects of 6 weeks' exposure to one of three diets known to produce obesity in rodents: a high-fat diet, a highly palatable "supermarket diet," and a normal chow diet supplemented with sugar water (75). Energy intake increased relative to baseline on the high-fat diet (as predicted by HOM) and sugar-supplemented diets, but there was evidence of weight gain only on the high-fat diet. Neither the EBM nor the CIM predicts failure of monkeys to gain weight despite increased energy intake in the form of sucrose, and the association of weight gain with increased fat intake is specifically contrary to the CIM. On the supermarket diet, assumed to be hyperpalatable, an increase in neither energy intake nor weight was observed, which is also contrary to both the EBM and CIM.

The above research on fructose supplementation (74) also demonstrated homeostatic regulation of intake, as is known for humans (69, 76), because monkeys compensated for the additional energy eaten in the sweetened beverages by reducing energy intake from other foods. Compensation was incomplete, however, with the monkeys ingesting approximately 30 kcal/day more than prior to the provision of the beverages. Incomplete compensation for variation in energy intake is likewise known for humans (77, 78). This is contrary to Jen's (75) results above, in which energy compensation was not observed for increased sucrose intake by middle-aged male macaques.

Some studies have examined the specific influences of energy, nutrients, and sensory properties on food selection and the control of feeding in RM. As in humans, high fat and/or carbohydrate contents in test foods are reported to be highly palatable and divert RM from nutritional reference values in a manner suggesting that they assigned value to specific nutrients rather than energy intake per se (79). Consistent with Bremer et al.'s (74) results above, RM regulate energy intake through compensating for gastric preloads of macronutrients by decreasing intake at subsequent feeding periods. Although some studies did not distinguish the effects of specific macronutrients, Jen et al. (80) showed that protein delivered directly into the stomachs of RM via cannula while feeding reduced calorie intake, whereas neither fat nor carbohydrate provision had the same effect. Hannah et al. (81) obtained similar results for gastric manipulations of dietary

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

protein content in RM. This is consistent with the important role of protein in intake regulation by humans and the evidence that protein dilution may play a causal role in influencing food and energy overconsumption (the PLH) (5, 73). Inconsistent with the PLH, however, is the observation that free-ranging RM in the Cayo Santiago research colony developed obesity when supplementary fed a high-protein (26%) commercial diet (68).

Given the hierarchical nature of RM social organization (see the section titled Rhesus Macaque Evolutionary and Social Ecology above), significant social effects on feeding and energy intake might be expected. Wilson et al. (82) found in communally housed females that socially subordinate RM had elevated markers of chronic stress and consistently ate more food and energy than more dominant individuals. This was due partly to subordinate monkeys continuing to eat through the night, whereas dominants ate only during the day. Interestingly, Johnston et al. (83) found the opposite for chronically stressed RM, which showed a 20% reduction in calorie intake during the 24 h prior to intrafamily aggression. Social stress, due, for example, to low socioeconomic status and food insecurity, has likewise been associated with eating (dys)regulation and obesity in humans (84–86).

Several experiments performed on RM are thus potentially relevant to models of human obesity, but no consistent picture has emerged as to which model, if any, is appropriate for this species. One possible reason, as noted previously, is that these experiments were not designed specifically to distinguish between obesity models. Second, many are compromised by small sample sizes and other aspects of experimental design, such as the use of liquid diet formulations, which, in humans, elicit different regulatory responses than solid diets (87, 88). Third, as discussed in the sections titled Insights from Nutritional Ecology and The Rhesus Macaque as a Model System for Human Nutrition below, determining their relevance for human nutrition would benefit from improved insight into the nutritional ecology of the species.

Caloric Restriction

Research on many species of animals has suggested that caloric restriction (CR) is associated with increased longevity and delayed onset of age-related disorders (89). For practical reasons, most studies have been done on short-lived laboratory animals such as yeast, nematode worms, fruit flies, and rodents. This raises the biomedically important but empirically challenging question of whether and how CR is likewise beneficial in humans. As a long-lived species with 93% genetic similarity and a life history strategy that in many respects resembles that of humans (90), RM is regarded as an important model for examining the translatability of CR research to humans (21, 61, 91).

Excepting a preliminary study (92) that has been criticized for statistical and other methodological reasons (93), two studies have examined the effects of CR on lifespan and health in RM. A study based at the University of Wisconsin (UW) found that age-related disease and mortality were lower in 38 monkeys that were calorically restricted from age 7–14 years than the 38 controls (94). The other, out of the National Institute on Aging (NIA) in Maryland, found only modest positive effects on health but no extension of lifespan in 57 calorically restricted monkeys compared with 64 controls (95).

There has been substantial discussion on why the benefits of dietary restriction were greater in the UW than the NIA study, and what this indicates about the relationship between diet and aging in RM (59, 90, 96–98). A plausible explanation concerns differences in the diets and feeding regimes used in the two studies. In the UW study, control monkeys were fed ad libitum, whereas in the NIA study the controls were provided a regulated amount of food designed to support a healthy body weight. Additionally, in the UW study, the diets were lower in protein, higher in fat, and more highly processed than the NIA diets (96, 99), all of which could result in excess

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

consumption (5). These differences have raised the question of whether the results of the UW study reflected the benefits of CR in the experimental group or the costs of overeating in the reference group (90, 97, 100).

As noted previously (97), from a biomedical perspective there may be no inconsistency in the results, because the two studies represent different dietary scenarios. However, these studies were performed not to evaluate the effects of overconsumption versus normal consumption but to examine in a species with a life cycle more comparable to humans whether reducing consumption below ~30% of normal intakes extends lifespan, as had been observed for various short-lived animals, such as nematode worms, fruit flies, fish, and mice (94, 95). This is a more fundamental aim, which links CR in RM, and by proxy potentially in humans, not only to biomedicine but also to the rich body of evolutionary theory and ecological research that has been developed to explain the links between dietary restriction and lifespan. As we discuss next, this is another subject to which nutritional ecology can contribute.

INSIGHTS FROM NUTRITIONAL ECOLOGY

RM clearly offer opportunities as a model system for advancing the understanding of human nutrition. Additionally, the vast amount of experimental research performed on this species can contribute to functional, ecological, and evolutionary aspects of primatology and the biological sciences more broadly. However, their contribution would benefit from broadening the context of RM research in two respects: first, through designing and interpreting studies against the ecological and evolutionary context of the species, and second, through explicit consideration of what kind of model system RM is intended to represent for a given research question. We next discuss these issues in turn, considering in this section the ecological and evolutionary context and in the following section some different categories of model systems that RM could represent.

In these considerations, we explore links between recent ecological/evolutionary research on RM performed within the field of nutritional ecology and aspects of the ecological, social, and biomedical research discussed in previous sections. Nutritional ecology is an integrative field that conceptualizes nutritional outcomes such as health and disease as emerging from the interactions of organisms and food environments (101, 102) and frequently operationalizes this framework using a multidimensional modeling approach called the nutritional geometry framework (103, 104). This framework has been applied in experimental and observational studies of humans (5), in observational studies of many species of wild primates (105, 106), and in preclinical studies of laboratory model organisms such as rodents and fruit flies (107, this volume). It thus provides a useful device for interrelating across different taxa and areas of research.

Dietary Diversity

Their pronounced ecological and dietary diversity set RM apart from most primates as potential models for ancestral human nutritional ecology, with both species considered broad dietary generalist omnivores with highly flexible and varied diets (19, 108). However, important questions remain regarding the diversity of RM diets and its relationship to humans. One question concerns the relationship between dietary diversity at the species, population, and individual levels. Another is the relationship between dietary generalism at the level of foods eaten versus nutrients consumed (109).

The diets of RM as a species are clearly diverse, but substantial variation also exists across populations in the nature and diversity of foods eaten. This raises the question of to what extent the broad diet of RM reflects individuals' biological capacity to subsist on diverse diets versus a species composed of diverse, relatively specialized populations or even individuals within populations. In

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

a wide-ranging analysis based on stable isotopes of herbivorous mammals from the Miocene to the present, DeSantis et al. (110) concluded that almost all herbivorous mammal species, regardless of dietary strategy, are composed of individuals that are relatively specialized compared with the species overall. If the broad species diet of RM is similarly composed of specialized individuals or populations, this could indicate that it is an emergent outcome of the species' widespread ecology rather than a result of individual dietary flexibility. Interestingly, a recent study of Pleistocene human paleoecology concluded that the trait that enabled *Homo sapiens* to establish across the globe when other hominin species failed reflects a "generalist specialist" niche in which subpopulations were specialized on different resource combinations (51). The ecological diversity of both RM and humans could thus be underpinned by the two species occupying a similar dietary niche. More ecological research is needed to address this in RM and to determine, if the species does occupy a generalist specialist niche, as seems likely, and the extent to which this is based on genetic differentiation between populations, genetic variation within populations, or individual and social learning.

An important but deceptively complex consideration for such research is how dietary generalism versus specialism is defined, whether applied to individuals, populations, or species. Most commonly, and as in the discussion above, the term dietary generalist is applied to animals that eat a wide range of foods, and in some contexts this is useful. But this range could be assessed along many dimensions, and depending on which is chosen, a specific animal might be classified differently. For this reason, Machovsky-Capuska et al. (109) used proportions-based nutritional geometry (103) to develop a multidimensional framework of the nutritional niche, aiming to capture the extent to which a species is generalist versus specialist across the different levels of the dietary hierarchy (4, 111)—nutrients, foods, and/or diets. For example, an animal might in the conventional sense be considered a dietary generalist because it eats many different foods, but if all these foods have similar nutrient composition, then it is also a "food composition specialist" (**Figure 1**). Alternatively, it might be capable of eating nutritionally disparate foods (food composition generalist) but does so specifically because this enables it to maintain a relatively constant dietary balance of macronutrients (macronutrient specialist) (112). Further, it might have limited physiological flexibility to tolerate variation in the intake of one nutrient, say, protein (protein specialist), but have metabolic flexibility in relation to fat and carbohydrates (**Figure 2*a*,*b***). Finally, foods also vary substantially in non-nutritional properties, such as the acquisition and processing adaptations required to exploit them. For example, as noted above, humans generally have a larger proportion of animal foods in their diets than RM, related to an ability to exploit a wider range of animal resources, including invertebrates and large mammals (113). They also have a less specialized gut structure than more strictly herbivorous primates such as RM, with a large intestine somewhat shorter relative to body size (more carnivore-like) compared with less omnivorous species (33). In this sense, humans can be considered "food exploitation generalists" (109).

The degree of flexibility an animal has at the various dietary levels is critical to understanding how diet constrains its ecological flexibility, and an area that requires more research attention. However, as we next discuss, nutritional geometry has enabled quantification of dimensions of dietary flexibility that are important when comparing humans and RM in biomedical and ecological research.

Obesity

In nutritional ecology, a key node for understanding obesity is the interface at which organismal biology interacts with food environments to generate excess intakes of energetic substrates that trigger fat storage (4). Comparative research applying nutritional geometry in experimental studies of nonhumans suggests that a near-universal mechanism for mediating the nutritional interactions of an organism with food environments is nutrient-specific appetites, whereby animals

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

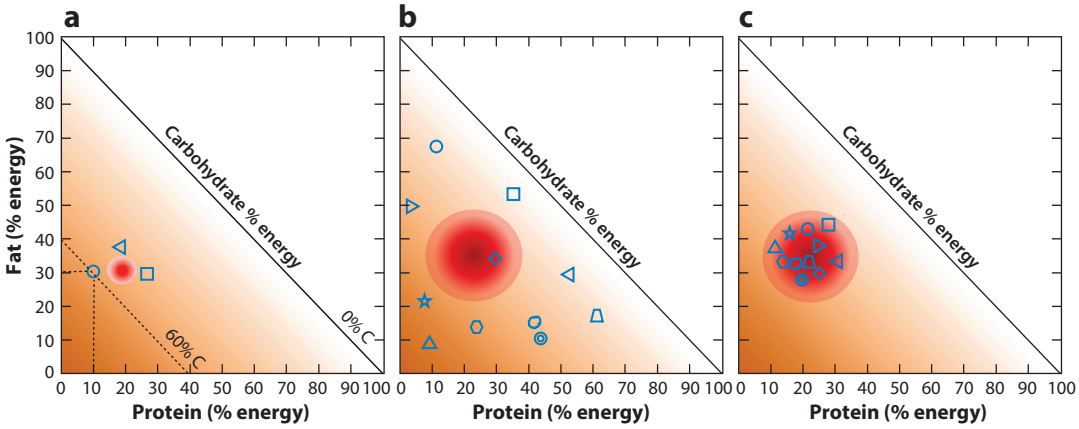


Figure 1

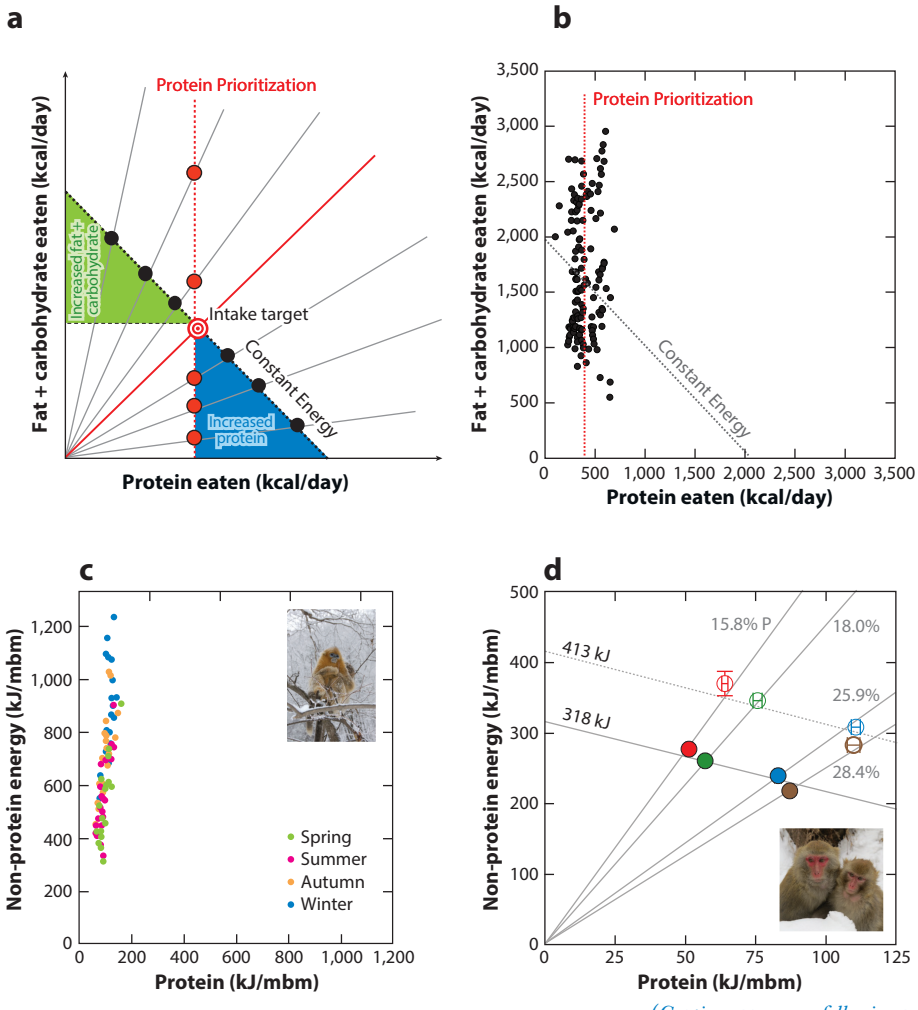
The multidimensional nutritional niche categorizes animals along the generalist–specialist spectrum at multiple levels using nutritional geometry. In this hypothetical illustration, right-angled mixture triangles (105) show the relationships between an animal's macronutritional requirements (*red graded area*) and the compositions of foods that are combined in the diet to meet its macronutrient needs (*blue symbols*). In these plots, the % energy contributed by protein (P) and fat (F) increase along the *x*- and *y*-axes, respectively, and the percentage of carbohydrate (C) increases along the color gradient from the diagonal marked 0% C to the origin (100% C) according to the formula %C = 100 – (%P + %F). Each point thus shows the 3D composition of a mixture (food or diet), as illustrated for the food denoted by the blue circle in panel *a* (10%P:30%F:60%C). (*a*) A hypothetical animal that eats few foods (a food specialist), eats foods that are similar in macronutrient composition (food composition specialist), and has narrow macronutrient requirements (*small red area* = macronutritional specialist). (*b*) An animal that is generalist at the levels of food (eats many foods), food composition (foods vary widely in composition), and macronutritional needs (adapted to tolerate a wide range of dietary compositions). (*c*) An animal that eats many foods (food generalist), eats foods that have similar macronutrient compositions (food composition specialist), and can tolerate wide variation in diet macronutrient composition (macronutritional generalist).

select a balanced complement of nutrients that support biological fitness (termed intake target) through homeostatic regulation of the consumption of different macronutrients (and some micronutrients) separately (107 in this volume). However, in imbalanced dietary situations where animals are constrained from achieving their target intake, nutrient-specific appetite systems are forced into a trade-off between over-ingesting some nutrients to compensate for dilution of the limiting ones and under-eating limiting nutrients to avoid surpluses of others. Specific strategies for resolving this trade-off (rules of compromise) have evolved in accordance with the biological, evolutionary, and ecological circumstances of different species (104, 114) (**Figure 2*a***).

Experimental and observational studies have shown that humans, too, have macronutrient-specific appetites and, when confined to macronutrient-imbalanced diets, prioritize achieving the target for protein most strongly, allowing food and energy intake to vary passively with variation in dietary protein concentration (5, 115) (**Figure 2*b***). Evidence suggests that over recent decades this pattern of intake regulation, termed Protein Prioritization, has interacted with increased consumption of protein-dilute diets, largely composed of ultraprocessed foods, to drive the epidemic of obesity and associated diseases [the PLH discussed in the previous section (5, 73)]. Observational field studies have shown that many species of nonhuman primates likewise show the Protein Prioritization pattern of macronutrient regulation (e.g., **Figure 2*c***), including two extant nonhuman hominids [chimpanzees (116) and orangutans (117)] and several species of new- and old-world monkeys (106). In the multidimensional niche framework discussed above, Protein Prioritization is a form of nutritional specialization that reflects limited tolerance of variation in protein intake (**Figure 2*a***).

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

Against this background, in a five-year observational study, Cui et al. (118, 119) tested whether RM in the Taihangshan Mountains of Northern China (**Figure 3**) likewise show Protein Prioritization, as might be expected given their aforementioned biological and ecological similarities with humans. Surprisingly, results showed that this was not the case (**Figure 2*d***). The study population maintained constant energy intake across successive springs in which the percentage of calories contributed by protein varied from 15.8% to 28.4%, a range approximately equal to that of normal human diets. Energy intake was maintained constant through increasing and decreasing calories from protein to offset ecological variation in the availability of nonprotein (fat and carbohydrate) energy. However, energy intakes were higher during the cold seasons (autumn and winter; data not shown) and for lactating females in spring (**Figure 2*d***), but equally so across years regardless of the wide variation in the percentage of energy contributed by protein. This suggests that energy intakes were regulated actively rather than passively following ecological variation in resource availability, thus reflecting homeostatic processes.



(*Caption appears on following page*)

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

Figure 2 (*Figure appears on preceding page*)

Nutritional geometry model showing two macronutrient regulatory strategies, built on axes depicting the amount of protein and nonprotein (fat + carbohydrate) energy eaten on diets varying in the ratio of protein:nonprotein energy. (*a*) A model in which the radial lines represent diets varying in the ratio of protein:fat + carbohydrate, with the red radial representing the optimally balanced (target) diet. The intake target represents an optimal intake point, and the red and black points are intakes under two different strategies of response to macronutrient imbalance. Under Protein Prioritization (*red*, *vertical*), protein intake is maintained constant, whereas fat + carbohydrate intake vary with dietary macronutrient balance. In the Constant Energy strategy, the macronutrients are weighted by the animal as interchangeable, such that on high protein:fat + carbohydrate diets, excess protein is eaten (*blue area*), and vice versa (*green area*), to maintain macronutrient energy intake constant. Animals that show Protein Prioritization are considered protein specialists because they show low tolerance of variation in protein intake relative to fat and carbohydrate. In contrast, the Constant Energy strategy represents greater flexibility in relation to dietary macronutrient composition. A physiological mechanism that would facilitate this flexibility is gluconeogenesis, in which glucose is liberated from excess ingested amino acids to supplement the shortfall of fat + carbohydrates when feeding on high-protein diets (*blue shaded area*). (*b*) Experimental (*shown here*) and observational studies demonstrate that humans show Protein Prioritization. Redrawn using data from Gosby et al. (151) and Raubenheimer et al. (152). (*c*) Golden snub-nosed monkeys, like humans and many other nonhuman primates, show Protein Prioritization. Data show daily individual intakes of semi-free-ranging captive monkeys within and across four seasons [modified from Hou et al. (121)]. (*d*) Unlike humans and most nonhuman primates studied, Taihangshan rhesus macaques maintained energy intake at near-constant levels across four successive springs in which the dietary percentage of energy from protein spanned 15.8–28.4%. Non-lactating adults (*solid symbols*) and lactating females (*hollow circles*) showed the same pattern, but energy intakes during lactation were substantially increased (413 versus 318 kJ), suggesting the patterns are driven by homeostatic regulation. Adapted from Cui et al. (118). Photo credits (*c*) D. Raubenheimer and (*d*) Z. Cui.

In a key respect related to obesity, therefore, this study population of RM is less like humans than most other primates studied to date: Their energy intake did not increase but was maintained constant, as dietary protein was diluted by carbohydrate and fat. One possible explanation for this difference is that the study population inhabits a temperate environment (**Figure 3*a*,*b***), whereas among other primate species Protein Prioritization appears to be associated with frugivory in tropical environments (120). This is not, however, a sufficient explanation, because golden snub-nosed monkeys inhabiting a similarly temperate environment show Protein Prioritization (121) (**Figure 2*c***). It thus remains to be determined whether the apparent absence of Protein Prioritization is a property of RM more generally, whether it is restricted to the highly temperate Taihangshan population studied by Cui et al. (119), or whether it is restricted to some populations but not others.

Although the answer to this question remains unknown, results of some existing studies are potentially relevant. Consistent with the hypothesis that the absence of Protein Prioritization observed in the Taihangshan RM is peculiar to this population, it is the only population of RM in which Liu et al. (41) found positive selection for genes involved in gluconeogenesis. As discussed above (see the section titled Rhesus Macaque Evolutionary and Social Ecology), gluconeogenesis is an important physiological trait for generating glucose from amino acids in animals adapted to high-protein diets, such as carnivores (122), and is increased in humans when confined to high-protein diets (123). On the other hand, free-ranging RM in the tropical Cayo Santiago research colony, which are not of temperate origin, developed obesity when supplemented by a high-protein (26%) commercial diet, which would not be expected if tropical RM showed Protein Prioritization. We are currently investigating these issues in comparative field studies of different RM populations.

In addition to the relevance of such studies for the PLH, another model of human obesity for which nutritional ecology studies of RM might provide insight is the FSH discussed above

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

Figure 3

Taihangshan rhesus macaques live in a highly temperate environment with (*a*) a mild spring and summer and (*b*) a harsh winter. Biological and temporal aging are clearly apparent in this wild population. Shown are (*c*) a 10-year-old alpha male and (*d*) his 32-year-old predecessor that he displaced 3 years prior. This could, additionally, illustrate nepotism, because the incumbent is the son of the older animal, and the transition was peaceful (Z. Cui et al., manuscript in preparation). Photo credits: (*a*,*c*,*d*) D. Raubenheimer and (*b*) Z. Cui.

(71, 124). The FSH considers fructose to be particularly important in obesity through triggering a survival mechanism that causes the depletion of intracellular ATP, which the body reads as a signal of impending crises and responds to by storing fat. This is considered an adaptive response in other species, but in humans to be overactivated and capable of causing obesity for two reasons. First, human diets are chronically rich in fructose, a sugar that in nature is found predominately in fruits and honey but in modern food environments is abundant in processed foods and beverages and is also endogenously synthesized following consumption of processed diets rich in high-glycemic sugars or salt. Second, in the middle Miocene, the great ape lineage from which humans evolved underwent a mutation that deactivated the uricase gene that degrades uric acid, which is a by-product of fructose metabolism and inhibits mitochondrial ATP synthesis. In the FSH, therefore, obesity results from ATP depletion due to a combination of oversupply of fructose and poor elimination of uric acid.

As noted above (see the section titled The Rhesus Macaque in Nutrition-Related Biomedical Research), RM can be useful models because, like humans, they develop obesity when experimentally exposed to high-energy, fructose-enriched diets (74). However, unlike humans, RM have functional uricase genes (125), raising the question of the extent to which uricase deactivation is a necessary mechanistic component of the explanation for obesity in RM and humans. Additionally, although many macaques are considered to feed predominately on fruits (24), the contribution of

fruit, and likely therefore also fructose, to the RM diet is highly variable but generally very low (45, 126). For example, whereas fruits made up as much as 79.4% of the diet in some populations (127), the diet of the population studied by Cui et al. (128) comprised just 1.4% fruit. Additionally, the diet of this population is composed largely of fibrous, presumably low-glycemic foods such as leaves and acorns, and there is no reason to suspect they have high sodium intake. Nonetheless, body weight in this population fluctuated across the year by as much as 10–15%.

Of the predominant obesity models, the PLH and FSH are the most overtly ecological and evolutionary, and thus more readily engage with nutritional ecology (from which the PLH was derived). It is, nonetheless, interesting to speculate on the relevance of ecological studies of RM to the EBM and CIM. Two important factors to consider are the adaptive benefits of fat storage and the details of the ecological vicissitudes that have selected for the mechanisms of energy and nutritional homeostasis under contention in these obesity models. From this perspective, considering the important role of reliable energy provision in the life cycle of primates, ecological generalist species like RM and humans would be predicted to evolve metabolic systems capable of opportunistically storing different energy substrates to meet future energy demands as permitted by available resources (consistent with the EBM), rather than evolving energy storage mechanisms that rely on a narrow dietary cue, such as high-glycemic sugars (the CIM). Although carbohydrates almost certainly are the dominant form of energy in the diets of most RM populations, as noted above (see the section titled Rhesus Macaque Evolutionary and Social Ecology), some also ingest significant amounts of lipid-rich foods, such as invertebrates. An intriguing possibility, following the generalist specialist dietary niche concept, is that the triggers and mechanisms of energy storage have been optimized for the specific ecological circumstances of different RM populations, such that their alignment with different models of obesity is similarly population dependent. Further studies on the relationships between ecological variation in food supply, dietary macronutrient mixtures, and body fat cycles across RM populations are needed to evaluate where they stand in relation to different models of human obesity, and to determine the extent to which experimental studies of captive macaques are relevant to human obesity.

Caloric Restriction

Nutritional ecology can likewise help to inform studies on the effects of CR on health and longevity in RM and to disentangle the evolved life-cycle responses of dietary restriction from pathological responses that are caused by feeding the control group diets to which primates are not adapted (discussed in the section titled The Rhesus Macaque in Nutrition-Related Biomedical Research).

The prevailing evolutionary hypothesis to explain the life-extending effects of CR is that it reflects an optimal resource-allocation strategy in which animals redirect metabolic resources away from reproduction to somatic maintenance during periods of food scarcity, thereby increasing the chances of surviving to reproduce when resource availability permits (129). However, accumulating evidence implicates the restriction not of calories per se, but specifically of protein and certain amino acids, as a proximate nutritional link between dietary restriction and lifespan (107, 130–132). Whether this is consistent with the resource-allocation theory depends on the extent to which protein (or specific amino acids) versus energy is more likely to have limited reproduction and somatic maintenance within the evolutionary environment of a species. Experimental studies have used nutritional geometry to address these issues through examining the relationships between dietary macronutrient balance, caloric intake, lifespan, and reproduction in laboratory model systems. Initially in fruit flies and other insects, and subsequently in mice, it was demonstrated that reproduction peaks at higher, and lifespan peaks at lower, dietary protein:carbohydrate ratios (reviewed in 107). As protein increased further, however, reproductive output decreased,

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

suggesting reproduction is maximized at an optimal rather than maximal proportion of energy from protein.

These results from laboratory model animals are potentially relevant to both biomedical and evolutionary research. In relation to biomedical research, a range of biological markers of health were measured in the mouse experiments, linking their responses to dietary macronutrient ratios and CR to human physiology (107 in this volume). Likewise, the benefits of low protein:carbohydrate diets in flies and mice are perhaps relevant to the exceptionally long health spans in human populations such as traditional Okinawans and other Blue Zone populations, who have diets on the lower end of the human range of protein:energy ratios (130). Ostensibly, this suggests the apparent paradox that low-protein diets might impact health both negatively through driving obesity (protein leverage, discussed above) and positively through extending health span. Research has shown, however, that the effects of protein dilution depend on the dietary carbohydrate:fat ratio and the type of carbohydrate involved (107, 133).

From an evolutionary perspective, the correspondence between low dietary protein: carbohydrate and increased lifespan in taxa as disparate as flies and mice (and possibly humans) suggests it might reflect a general biological pattern, which could point to a new evolutionary understanding of the links between diet and lifespan. Specifically, it might suggest that the association is driven not by a trade-off in allocation of scarce resources between lifespan and reproduction, as postulated in resource-allocation theories, but by incompatible dietary optima in which animals are adapted to select higher-protein diets that optimize reproduction but are physiologically incompatible with extended lifespan (134).

Because aging can be observed in the wild (135) for many species, including RM (**Figure 3*c*,*d***), field studies can help to better understand the relationships between diet, reproduction, health, and lifespan in an ecological context. It would, for example, be interesting to know what constitute optimal diets versus restricted diets, and whether long periods during which the Taihangshan macaques are ecologically constrained to high-protein diets (118, 119) have life-shortening effects compared with populations not restricted in this way. Such studies would help elucidate how patterns of scarcity (e.g., protein versus calories) influence reproduction and how to design "normal" diets relative to different restriction treatments in laboratory studies of model animals such as the NIA and UW RM experiments discussed above.

Finally, in addition to dietary quality, another context in which field studies of RM could inform biomedical research on the relationship between diet and longevity is the temporal pattern of feeding. In modern humans, most food intake is gathered in structured meal-taking periods, typically in the morning, midday, and evening (136). In rodent model systems, the circadian pattern of eating can significantly influence health and lifespan (137, 138), and evidence suggests the same might be true for humans (139, 140). Few studies of which we are aware have systematically documented the temporal pattern of feeding in wild RM, but our own studies indicate that feeding occurs throughout the day, with peaks in the number of animals feeding (though not necessarily the amount eaten by individuals) between approximately 09:00–10:00 and 14:00–15:00 (141). It would be informative to better understand how the natural pattern of feeding varies within and across RM populations, its relation to variability in human patterns of feeding in industrialized societies (136), and how this relates to evolutionary and biomedical aspects of health and lifespan.

THE RHESUS MACAQUE AS A MODEL SYSTEM FOR HUMAN NUTRITION

The above discussions provide some interesting perspectives to evaluate the utility of RM as model organisms for human nutrition by considering how they are chosen for distinct purposes. In biomedical research, model organisms serve primarily as a tractable system in which to study a

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

more complex or inaccessible phenomenon in humans (142, 143). In this regard, RM serve as *surrogate* models by standing in for a target, e.g., humans, usually due to practical or ethical constraints with studying the target itself (144). To be a useful surrogate, the model need not be phylogenetically related to the target, because the aim is to investigate functional and mechanistic similarities, for example, a specific pathway or mechanism that responds similarly to manipulations as the target, regardless of relatedness. Surrogates thus serve as potentially powerful tools to unravel the etiology and mechanisms of human disease.

As we saw, RM are sometimes used as surrogates for human nutrition given that both are dietary generalist omnivores and respond similarly to dietary interventions; e.g., both captive and free-ranging populations of RM can develop spontaneous obesity. Moreover, their genetic, physiological, and life history similarities to humans support the role of RM as potential surrogates to test the effects of caloric and dietary restriction. However, as nutritional ecology studies have shown, at least one population of RM does not respond to nutritional variations in the same way as humans (some RM populations prioritize energy, versus Protein Prioritization in humans). This nuances their status as surrogates for studying food selection, disease, and aging outcomes because their "normal" nutritional priorities differ. Furthermore, this may be population dependent, demonstrating the value for biomedical research to consider ecological variation among RM populations (41).

In contrast to surrogate models in biomedical research, within basic biological research, model organisms such as *Drosophila melanogaster*, *Caenorhabditis elegans*, mice, and sometimes macaques are chosen not for their ability to represent a particular target species but to represent multiple species and to thus uncover widely, or even universally, shared biological phenomena (144). In this sense, models can serve as exemplars of increasingly wide "representational scope" (145), helping to reveal conserved traits or mechanisms across populations, species, and taxa. To maximize the generalization potential of exemplars, it is usually best to avoid organisms or populations with unique features or that are outliers (144). It is also common to study these organisms in simplified laboratory environments, largely setting aside the ecological and evolutionary context (146).

By helping to isolate and understand the genetics of obesity and diabetes and the various peptides and hormones regulating organismal feeding and dietary regulation, as well as further illustrating how nutrient-specific appetites are basic (and highly conserved) biological traits, RM play key roles as exemplar models. Similar to their potential limitations as surrogates for studying obesity, however, whether humans fall within the representational scope of RM as exemplars will depend on the trait under investigation and possibly on the population being studied, some of which may be genetic and behavioral outliers.

Although standard model organisms enable generalizations to humans based on shared underlying traits, thus supporting their "translational power" in biomedical research (20, 147), there are also benefits to studying how models differ from humans, thus serving as "negative models" (148). Studying how, for example, desert rodents have adapted to the extremes of water scarcity by evolving very long loops of Henle (149) can serve as a useful contrast case to humans or other mammals. Because such organisms display distinct adaptations to relatively extreme environments, this contrast can foster understanding of a trait's biological diversity, and it may elucidate how other organisms adapt to similar, yet less extreme, environmental constraints. By uncovering diversity, negative models can thereby help expose blind spots that might come from studying standardized models, e.g., rodents (148), and the comparative method of studying diverse adaptations may lead to general or unifying principles.

This indicates an unexplored potential for viewing RM as negative models, because in some respects relevant to nutrition they are dissimilar to humans. For example, although RM lack the relevant uricase mutation found in humans and consume varying levels of fructose, their

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

dissimilarity might prove useful, e.g., to test the extent to which this mutation is necessary for obesity explanations in humans (as predicted by the FSH). Likewise, regarding protein leverage, at least some populations of RM do not prioritize protein and thus serve as an interesting contrast case, while still illustrating the broader principles of nutrient balancing.

The potential utility of RM as model systems thus comes from aspects that are specifically like humans (surrogate models), aspects that are shared more broadly across species (exemplars), and potentially aspects that are specifically different from humans (negative models). Explicit consideration of which model category applies in specific uses of RM could benefit biomedical research. For example, given the ethical and logistical challenges in the use of captive nonhuman primate models, is it justifiable to use RM as exemplar models to examine broadly generalizable traits that could equally be studied in other models such as rodents or insects? Likewise, the implications for human nutritional pathophysiology of experimental dietary macronutrient manipulation could be fundamentally different for a population of RM that shows Protein Prioritization (a surrogate model) versus one that eats to Constant Energy (a negative model). These considerations highlight different ways in which RM could feature prominently as nutritional models in biomedicine and basic biology, and how ecological and evolutionary research can help inform this.

Exciting prospects lie also in the potential for RM to act as a model for human nutritional ecology. For this, it is worth briefly contrasting how model systems are defined and used in evolutionary and ecological research (EE) versus laboratory-based experimental research. In contrast to basic research, where models usually are selected to represent various species (i.e., as exemplars with wide representational scope), and to biomedical research focusing on the mechanistic similarities to a single species, usually humans, the function of model organisms in EE differs in several respects (150). First, in EE, model organisms serve principally to understand genetic and phenotypic variation among organisms, thus supporting comparative research (in this respect similar to negative models). Second, this variation is understood in the context of the natural environment of the model, and crucially, the environment is part of the system being studied (150), whereas in basic and biomedical research the environment is usually not included (146). Third, in EE, model organisms are situated in a continuum of "model systems" that spans levels of biological organization within and beyond the individual, including cells, populations, ecological communities, and ecosystems. Finally, whereas the emphasis in basic and biomedical research is on the amenability of the model organism for laboratory research, in EE the relevant setting could equally be field studies of the model system (organism, population, community, or ecosystem).

Future research can help nuance the role of RM as biomedical surrogates, for human nutritional ecology, or as exemplars more broadly (e.g., of primates in general) by expanding or redrawing the boundaries of the model "system." First, the RM model should explicitly include relevant ecological factors, such as food type and availability, climate, and social interactions. Second, it should embrace the diversity in these relationships both within and across populations to reflect how genetic variations interact with physiological, behavioral, and cultural traits to produce the diverse nutritional strategies of RM. In this regard, RM could be the ultimate surrogate for studying human dietary and nutritional adaptation, including the diversity of human populations and their ecological interactions with increasingly global obesogenic food environments (4). Simultaneously, the prodigious adaptability of RM could provide a negative model to help understand and manage the conservation crisis in relation to most primates and other taxa that struggle to adapt to the rapidly changing world.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

ACKNOWLEDGMENTS

We thank the *Annual Review of Animal Biosciences* for the invitation to contribute this review. The work was supported by the National Natural Science Foundation of China (32370520, 32070446), China Postdoctoral Science Foundation (2019M662520), Top-notch Talents of Zhengzhou University (32211502, 32212341, 32340114, 32340438, 35220132), and the Australian National Health and Medical Research Council Program Grant (GNT1149976). Thanks to Paul Griffiths for his generous contribution through his Laureate Fellowship, A Philosophy of Medicine for the 21st Century (FL170100160), which enabled the collaboration of Jonathan Sholl and David Raubenheimer.

LITERATURE CITED

1. Afshin A, Sur PJ, Fay KA, Cornaby L, Ferrara G, et al. 2019. Health effects of dietary risks in 195 countries, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017. *Lancet* 393(10184):1958–72
2. Brown AW, Aslibekyan S, Bier D, Ferreira da Silva R, Hoover A, et al. 2023. Toward more rigorous and informative nutritional epidemiology: the rational space between dismissal and defense of the status quo. *Crit. Rev. Food Sci. Nutr.* 63(18):3150–67
3. Flanagan A, Bradfield J, Kohlmeier M, Ray S. 2023. Need for a nutrition-specific scientific paradigm for research quality improvement. *BMJ Nutr. Prev. Health* 6(2):383–91
4. Raubenheimer D, Simpson SJ. 2016. Nutritional ecology and human health. *Annu. Rev. Nutr.* 36:603–26
5. Raubenheimer D, Simpson SJ. 2023. Protein appetite as an integrator in the obesity system: the protein leverage hypothesis. *Philos. Trans. R. Soc. B* 378(1888):20220212
6. Domínguez-Oliva A, Hernández-Ávalos I, Martínez-Burnes J, Olmos-Hernández A, Verduzco-Mendoza A, Mota-Rojas D. 2023. The importance of animal models in biomedical research: current insights and applications. *Animals* 13(7):1223
7. Conn PM, ed. 2017. *Animal Models for the Study of Human Disease*. Cambridge, MA: Academic. 2nd ed.
8. Sterelny K. 2017. Humans as model organisms. *Proc. R. Soc. B* 284(1869):20172115
9. Rubio-Aliaga I. 2012. Model organisms in molecular nutrition research. *Mol. Nutr. Food Res.* 56(6):844–53
10. Baker DH. 2008. Animal models in nutrition research. *J. Nutr.* 138(2):391–96
11. Katz PS. 2016. 'Model organisms' in the light of evolution. *Curr. Biol.* 26(14):R649–50
12. Ankeny RA, Leonelli S. 2020. *Model Organisms*. Cambridge Elem. Philos. Biol. Cambridge, UK: Cambridge Univ. Press
13. Eur. Parl. 2021. *Plans and actions to accelerate a transition to innovation without the use of animals in research, regulatory testing and education*. Text adopt., P9_TA(2021)0387, Eur. Parl., Strasbourg, Fr.
14. Hunter P. 2008. The paradox of model organisms. The use of model organisms in research will continue despite their shortcomings. *EMBO Rep.* 9(8):717–20
15. Kiani AK, Pheby D, Henehan G, Brown R, Sieving P, et al. 2022. Ethical considerations regarding animal experimentation. *J. Prev. Med. Hyg.* 63(2 Suppl. 3):E255–66
16. Striedter GF. 2019. Variation across species and levels: implications for model species research. *Brain Behav. Evol.* 93(2–3):57–69
17. Phillips KA, Bales KL, Capitanio JP, Conley A, Czoty PW, et al. 2014. Why primate models matter. *Am. J. Primatol.* 76(9):801–27
18. Xue C, Raveendran M, Harris RA, Fawcett GL, Liu X, et al. 2016. The population genomics of rhesus macaques (*Macaca mulatta*) based on whole-genome sequences. *Genome Res.* 26(12):1651–62
19. Cooper EB, Brent LJ, Snyder-Mackler N, Singh M, Sengupta A, et al. 2022. The rhesus macaque as a success story of the Anthropocene. *eLife* 11:e78169
20. Tarantal AF, Noctor SC, Hartigan-O'Connor DJ. 2022. Nonhuman primates in translational research. *Annu. Rev. Anim. Biosci.* 10:441–68
21. Colman RJ. 2018. Non-human primates as a model for aging. *Biochim. Biophys. Acta Mol. Basis Dis.* 1864(9 Pt. A):2733–41

22. Disotell TR, Tosi AJ. 2007. The monkey's perspective. *Genome Biol*. 8(9):226
23. Zhang B-L, Chen W, Wang Z, Pang W, Luo M-T, et al. 2023. Comparative genomics reveals the hybrid origin of a macaque group. *Sci. Adv.* 9(22):eadd3580
24. Roos C, Zinner D. 2015. Diversity and evolutionary history of macaques with special focus on *Macaca mulatta* and *Macaca fascicularis*. In *The Nonhuman Primate in Nonclinical Drug Development and Safety Assessment*, ed. J Bluemel, S Korte, E Schenck, GF Weinbauer, pp. 3–16. San Diego, CA: Academic
25. Li J, Han K, Xing J, Kim H-S, Rogers J, et al. 2009. Phylogeny of the macaques (Cercopithecidae: *Macaca*) based on *Alu* elements. *Gene* 448(2):242–49
26. Richard AF, Goldstein SJ, Dewar RE. 1989. Weed macaques: the evolutionary implications of macaque feeding ecology. *Int. J. Primatol.* 10(6):569–94
27. Meindl RS, Chaney ME, Lovejoy CO. 2018. Early hominids may have been weed species. *PNAS* 115(6):1244–49
28. Sengupta A, Radhakrishna S. 2018. The hand that feeds the monkey: mutual influence of humans and rhesus macaques (*Macaca mulatta*) in the context of provisioning. *Int. J. Primatol.* 39(5):817–30
29. Anderson CJ, Johnson SA, Hostetler ME, Summers MG. 2016. *History and status of introduced rhesus macaques (*Macaca mulatta*) in Silver Springs State Park*, *Florida*. Publ. WEC367, Dep. Wildl. Ecol. Conserv., Univ. Fla., Gainesville
30. Evans BJ, Gansauge M-T, Tocheri MW, Schillaci MA, Sutikna T, et al. 2020. Mitogenomics of macaques (*Macaca*) across Wallace's Line in the context of modern human dispersals. *J. Hum. Evol.* 146:102852
31. Leonard WR, Stock JT, Valeggia CR. 2010. Evolutionary perspectives on human diet and nutrition. *Evol. Anthropol.* 19(3):85–86
32. Sarker G, Kabir M, Feeroz M, Hasan M. 2008. Food and feeding behaviour of rhesus macaque (*Macaca mulatta*) at Barmi, Gazipur, Bangladesh. *J. Life Sci.* 20:1–8
33. Pontzer H, Wood BM. 2021. Effects of evolution, ecology, and economy on human diet: insights from hunter-gatherers and other small-scale societies. *Annu. Rev. Nutr*. 41:363–85
34. Yuan Q, Zhou Z, Lindell SG, Higley JD, Ferguson B, et al. 2012. The rhesus macaque is three times as diverse but more closely equivalent in damaging coding variation as compared to the human. *BMC Genet*. 13(1):52
35. Warren WC, Harris RA, Haukness M, Fiddes IT, Murali SC, et al. 2020. Sequence diversity analyses of an improved rhesus macaque genome enhance its biomedical utility. *Science* 370(6523):eabc6617
36. Smith DG, McDonough J. 2005. Mitochondrial DNA variation in Chinese and Indian rhesus macaques (*Macaca mulatta*). *Am. J. Primatol.* 65(1):1–25
37. Fooden J. 2000. *Systematic Review of the Rhesus Macaque*, Macaca mulatta *(Zimmermann*, *1780)*. Chicago: Field Mus. Nat. Hist.
38. Groves C. 2001. *Primate Taxonomy*. Washington, DC: Smithson. Books. 1st ed.
39. Hernandez RD, Hubisz MJ, Wheeler DA, Smith DG, Ferguson B, et al. 2007. Demographic histories and patterns of linkage disequilibrium in Chinese and Indian rhesus macaques. *Science* 316(5822):240–43
40. Smith DG. 2005. Genetic characterization of Indian-origin and Chinese-origin rhesus macaques (*Macaca mulatta*). *Comp. Med.* 55(3):227–30
41. Liu Z, Tan X, Orozco-terWengel P, Zhou X, Zhang L, et al. 2018. Population genomics of wild Chinese rhesus macaques reveals a dynamic demographic history and local adaptation, with implications for biomedical research. *GigaScience* 7(9):giy106
42. Zhou Y, Tian J, Jiang H, Han M, Wang Y, Lu J. 2024. Phylogeography and demographic history of macaques, fascicularis species group, in East Asia: inferred from multiple genomic markers. *Mol. Phylogenet. Evol*. 194:108042
43. Francis G, Wang Q. 2023. Coming to the Caribbean—acclimation of Rhesus macaques (*Macaca mulatta*) at Cayo Santiago. *Am. J. Biol. Anthropol.* 181(2):271–95
44. Pathak AK, Sukhavasi K, Marnetto D, Chaubey G, Pandey AK. 2022. Human population genomics approach in food metabolism. In *Future Foods*, ed. R Bhat, pp. 433–49. Cambridge, MA: Academic
45. Thierry B. 2007. Unity in diversity: lessons from macaque societies. *Evol. Anthropol.* 16(6):224–38
46. Adams MJ, Majolo B, Ostner J, Schülke O, De Marco A, et al. 2015. Personality structure and social style in macaques. *J. Personal. Soc. Psychol.* 109(2):338–53

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

47. Thierry B. 2000. Covariation of conflict management patterns across macaque species. In *Natural Conflict Resolution*, pp. 106–28. Berkeley: Univ. Calif. Press
48. Izumiyama S. 2014. Seasonal and altitudinal migration of Japanese Macaques in the Northern Japan Alps. In *High Altitude Primates*, ed. NB Grow, S Gursky-Doyen, A Krzton, pp. 153–81. New York: Springer
49. Fiore AM, Cronin KA, Ross SR, Hopper LM. 2020. Food cleaning by Japanese macaques: Innate, innovative or cultural? *Folia Primatol.* 91(4):433–44
50. Maestripieri D. 2007. *Macachiavellian Intelligence: How Rhesus Macaques and Humans Have Conquered the World*. Chicago: Univ. Chicago Press
51. Roberts P, Stewart BA. 2018. Defining the 'generalist specialist' niche for Pleistocene Homo sapiens. *Nat. Hum. Behav.* 2(8):542–50
52. Dubuc C, Hughes KD, Cascio J, Santos LR. 2012. Social tolerance in a despotic primate: co-feeding between consortship partners in rhesus macaques. *Am. J. Phys. Anthropol.* 148(1):73–80
53. Mazumder J, Kaburu SSK. 2021. First report of food sharing among nicobar long-tailed macaques. *Quat. Int.* 603:31–39
54. Amici F, Widdig A, MacIntosh AJJ, Francés VB, Castellano-Navarro A, et al. 2020. Dominance style only partially predicts differences in neophobia and social tolerance over food in four macaque species. *Sci. Rep.* 10(1):22069
55. Balasubramaniam KN, Marty PR, Samartino S, Sobrino A, Gill T, et al. 2020. Impact of individual demographic and social factors on human-wildlife interactions: a comparative study of three macaque species. *Sci. Rep.* 10(1):21991
56. Testard C, Larson SM, Watowich MM, Kaplinsky CH, Bernau A, et al. 2021. Rhesus macaques build new social connections after a natural disaster. *Curr. Biol.* 31(11):2299–309.e7
57. Testard C, Shergold C, Acevedo-Ithier A, Hart J, Bernau A, et al. 2024. Ecological disturbance alters the adaptive benefits of social ties. *Science* 384(6702):1330–35
58. Turner BL, Thompson AL. 2013. Beyond the Paleolithic prescription: incorporating diversity and flexibility in the study of human diet evolution. *Nutr. Rev.* 71(8):501–10
59. Havel PJ, Kievit P, Comuzzie AG, Bremer AA. 2017. Use and importance of nonhuman primates in metabolic disease research: current state of the field. *ILAR J.* 58(2):251–68
60. Kleinert M, Clemmensen C, Hofmann SM, Moore MC, Renner S, et al. 2018. Animal models of obesity and diabetes mellitus. *Nat. Rev. Endocrinol.* 14(3):140–62
61. Wood EK, Sullivan EL. 2022. The influence of diet on metabolism and health across the lifespan in nonhuman primates. *Curr. Opin. Endocr. Metab. Res.* 24:100336
62. Kemnitz JW. 1984. Obesity in macaques: spontaneous and induced. *Adv. Vet. Sci. Comp. Med.* 28:81–114
63. Bauer SA, Arndt TP, Leslie KE, Pearl DL, Turner PV. 2011. Obesity in rhesus and cynomolgus macaques: a comparative review of the condition and its implications for research. *Comp. Med.* 61(6):514–26
64. Mattison JA, Vaughan KL. 2017. An overview of nonhuman primates in aging research. *Exp. Gerontol.* 94:41–45
65. Kemnitz JW, Francken GA. 1986. Characteristics of spontaneous obesity in male rhesus monkeys. *Physiol. Behav.* 38(4):477–83
66. Vaughan KL, Mattison JA. 2016. Obesity and aging in humans and nonhuman primates: a mini-review. *Gerontology* 62(6):611–17
67. West DB, York B. 1998. Dietary fat, genetic predisposition, and obesity: lessons from animal models. *Am. J. Clin. Nutr.* 67(Suppl. 3):505S–12S
68. Schwartz SM, Kemnitz JW. 1992. Age- and gender-related changes in body size, adiposity, and endocrine and metabolic parameters in free-ranging rhesus macaques. *Am. J. Phys. Anthropol.* 89(1):109–21
69. Hall KD, Farooqi IS, Friedman JM, Klein S, Loos RJF, et al. 2022. The energy balance model of obesity: beyond calories in, calories out. *Am. J. Clin. Nutr.* 115(5):1243–54
70. Ludwig DS, Ebbeling CB. 2018. The carbohydrate-insulin model of obesity: beyond "calories in, calories out." *JAMA Intern. Med.* 178(8):1098–103
71. Johnson RJ, Lanaspa MA, Sanchez-Lozada LG, Tolan D, Nakagawa T, et al. 2023. The fructose survival hypothesis for obesity. *Philos. Trans. R. Soc. B* 378(1885):20220230

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

72. Gao L, Hu S, Yang D, Wang L, Togo J, et al. 2024. The hedonic overdrive model best explains high-fat diet-induced obesity in C57BL/6 mice. *Obesity* 32(4):733–42
73. Simpson SJ, Raubenheimer D. 2005. Obesity: the protein leverage hypothesis. *Obes. Rev.* 6(2):133–42
74. Bremer AA, Stanhope KL, Graham JL, Cummings BP, Wang W, et al. 2011. Fructose-fed rhesus monkeys: a nonhuman primate model of insulin resistance, metabolic syndrome, and type 2 diabetes. *Clin. Transl. Sci.* 4(4):243–52
75. Jen K-LC. 1987. Failure to induce weight gain with palatable diets in monkeys (*Macaca mulatta*). *Primates* 28(1):61–69
76. Hopkins M, Beaulieu K, Gibbons C, Halford JCG, Blundell J, et al. 2000. The control of food intake in humans. In *Endotext*, ed. KR Feingold, B Anawalt, MR Blackman, A Boyce, G Chrousos, et al. South Dartmouth, MA: MDText.com, Inc.
77. Levitsky DA, Sewall A, Zhong Y, Barre L, Shoen S, et al. 2019. Quantifying the imprecision of energy intake of humans to compensate for imposed energetic errors: a challenge to the physiological control of human food intake. *Appetite* 133:337–43
78. Almiron-Roig E, Palla L, Guest K, Ricchiuti C, Vint N, et al. 2013. Factors that determine energy compensation: a systematic review of preload studies. *Nutr. Rev.* 71(7):458–73
79. Huang F-Y, Sutcliffe MPF, Grabenhorst F. 2021. Preferences for nutrients and sensory food qualities identify biological sources of economic values in monkeys. *PNAS* 118(26):e2101954118
80. Jen K-LC, Bodkin NL, Metzger BL, Hansen BC. 1985. Nutrient composition: effects on appetite in monkeys with oral factors held constant. *Physiol. Behav.* 34(5):655–59
81. Hannah JS, Dubey AK, Hansen BC. 1990. Postingestional effects of a high-protein diet on the regulation of food intake in monkeys. *Am. J. Clin. Nutr.* 52(2):320–25
82. Wilson ME, Fisher J, Fischer A, Lee V, Harris RB, Bartness TJ. 2008. Quantifying food intake in socially housed monkeys: social status effects on caloric consumption. *Physiol. Behav.* 94(4):586–94
83. Johnston JR, Meeker TL, Ramsey JK, Crane MM, Cohen JK, Ethun KF. 2020. Utility of automated feeding data to detect social instability in a captive breeding colony of rhesus macaques (*Macaca mulatta*): a case study of intrafamily aggression. *J. Am. Assoc. Lab. Anim. Sci.* 59(1):46–57
84. Bateson M, Pepper GV. 2023. Food insecurity as a cause of adiposity: evolutionary and mechanistic hypotheses. *Philos. Trans. R. Soc. B* 378(1888):20220228
85. Scott KA, Melhorn SJ, Sakai RR. 2012. Effects of chronic social stress on obesity. *Curr. Obes. Rep.* 1(1):16–25
86. Hemmingsson E, Nowicka P, Ulijaszek S, Sørensen TIA. 2023. The social origins of obesity within and across generations. *Obes. Rev.* 24(1):e13514
87. Tieken SM, Leidy HJ, Stull AJ, Mattes RD, Schuster RA, Campbell WW. 2007. Effects of solid versus liquid meal-replacement products of similar energy content on hunger, satiety, and appetite-regulating hormones in older adults. *Horm. Metab. Res.* 39(5):389–94
88. Zheng M, Allman-Farinelli M, Heitmann BL, Toelle B, Marks G, et al. 2015. Liquid versus solid energy intake in relation to body composition among Australian children. *J. Hum. Nutr. Diet.* 28(Suppl. 2):70–79
89. Colman RJ, Anderson RM. 2011. Nonhuman primate calorie restriction. *Antioxid. Redox Signal.* 14(2):229–39
90. Le Bourg E. 2018. Does calorie restriction in primates increase lifespan? Revisiting studies on macaques (*Macaca mulatta*) and mouse lemurs (*Microcebus murinus*). *BioEssays* 40(10):e1800111
91. Chiou KL, Montague MJ, Goldman EA, Watowich MM, Sams SN, et al. 2020. Rhesus macaques as a tractable physiological model of human ageing. *Philos. Trans. R. Soc. B* 375(1811):20190612
92. Bodkin NL, Alexander TM, Ortmeyer HK, Johnson E, Hansen BC. 2003. Mortality and morbidity in laboratory-maintained Rhesus monkeys and effects of long-term dietary restriction. *J. Gerontol. A* 58(3):212–19
93. Lane MA, Mattison JA, Roth GS, Brant LJ, Ingram DK. 2004. Effects of long-term diet restriction on aging and longevity in primates remain uncertain. *J. Gerontol. A* 59(5):405–7
94. Colman RJ, Beasley TM, Kemnitz JW, Johnson SC, Weindruch R, Anderson RM. 2014. Caloric restriction reduces age-related and all-cause mortality in rhesus monkeys. *Nat. Commun.* 5:3557
95. Mattison JA, Roth GS, Beasley TM, Tilmont EM, Handy AM, et al. 2012. Impact of caloric restriction on health and survival in rhesus monkeys from the NIA study. *Nature* 489(7415):318–21

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

96. Mattison JA, Colman RJ, Beasley TM, Allison DB, Kemnitz JW, et al. 2017. Caloric restriction improves health and survival of rhesus monkeys. *Nat. Commun.* 8:14063
97. Austad SN, Hoffman JM. 2021. Beyond calorie restriction: aging as a biological target for nutrient therapies. *Curr. Opin. Biotechnol.* 70:56–60
98. Austad SN. 2012. Ageing: mixed results for dieting monkeys. *Nature* 489(7415):210–11
99. Rizza W, Veronese N, Fontana L. 2014. What are the roles of calorie restriction and diet quality in promoting healthy longevity? *Ageing Res. Rev.* 13:38–45
100. Sohal RS, Forster MJ. 2014. Caloric restriction and the aging process: a critique. *Free Radic. Biol. Med.* 73:366–82
101. Raubenheimer D, Simpson SJ, Tait AH. 2012. Match and mismatch: conservation physiology, nutritional ecology and the timescales of biological adaptation. *Philos. Trans. R. Soc. B* 367(1596):1628–46
102. Raubenheimer D, Simpson SJ, Mayntz D. 2009. Nutrition, ecology and nutritional ecology: toward an integrated framework. *Funct. Ecol.* 23(1):4–16
103. Raubenheimer D. 2011. Toward a quantitative nutritional ecology: the right-angled mixture triangle. *Ecol. Monogr.* 81(3):407–27
104. Simpson SJ, Raubenheimer D. 2012. *The Nature of Nutrition*. Princeton, NJ: Princeton Univ. Press
105. Lambert JE, Rothman JM. 2015. Fallback foods, optimal diets, and nutritional targets: primate responses to varying food availability and quality. *Annu. Rev. Anthropol.* 44:493–512
106. Raubenheimer D, Hou R, Dong Y, Ren C, Cui Z. 2023. Towards an integrated understanding of dietary phenotypes. *Philos. Trans. R. Soc. B* 378(1891):20220545
107. Senior AM, Raubenheimer D, Le Couteur DG, Simpson SJ. 2025. The geometric framework for nutrition and its application to rodent models. *Annu. Rev. Anim. Biosci.* 13:389–410
108. Zhang K, Karim F, Jin Z, Xiao H, Yao Y, et al. 2022. Diet and feeding behavior of a group of high-altitude rhesus macaques: high adaptation to food shortages and seasonal fluctuations. *Curr. Zool.* 69(3):304–14
109. Machovsky-Capuska GE, Senior AM, Simpson SJ, Raubenheimer D. 2016. The multidimensional nutritional niche. *Trends Ecol. Evol.* 31(5):355–65
110. DeSantis LRG, Pardi MI, Du A, Greshko MA, Yann LT, et al. 2022. Global long-term stability of individual dietary specialization in herbivorous mammals. *Proc. R. Soc. B* 289(1968):20211839
111. Sholl J, Raubenheimer D. 2023. Who's afraid of nutritionism? *Br. J. Philos. Sci.* **https://doi.org/10.1086/728785**
112. Raubenheimer D, Simpson SJ. 1999. Integrating nutrition: a geometrical approach. In *Proceedings of the 10th International Symposium on Insect-Plant Relationships*, ed. SJ Simpson, AJ Mordue, J Hardie, pp. 67–82. Dordrecht: Springer Neth.
113. Raubenheimer D, Rothman JM. 2013. Nutritional ecology of entomophagy in humans and other primates. *Annu. Rev. Entomol.* 58:141–60
114. Raubenheimer D, Simpson SJ. 1997. Integrative models of nutrient balancing: application to insects and vertebrates. *Nutr. Res. Rev.* 10(1):151–79
115. Gosby AK, Conigrave AD, Lau NS, Iglesias MA, Hall RM, et al. 2011. Testing protein leverage in lean humans: a randomised controlled experimental study. *PLOS ONE* 6(10):e25929
116. Uwimbabazi M, Raubenheimer D, Tweheyo M, Basuta GI, Conklin-Brittain NL, et al. 2021. Nutritional geometry of female chimpanzees (*Pan troglodytes*). *Am. J. Primatol.* 83(7):e23269
117. *Am. J. Phys. Anthropol.* Abstracts - AAPA presentations. 2015. *Am. J. Phys. Anthropol.* 156(S60):65–334
118. Cui Z-W, Zhang Y, Yan J-B, Zhang Y-F, Dong Y-L, et al. 2022. What does it mean to be a macronutritional generalist? A five-year case study in wild rhesus macaques (*Macaca mulatta*). *Zool. Res.* 43(6):935–39
119. Cui Z-W, Wang Z-L, Shao Q, Raubenheimer D, Lu J-Q. 2018. Macronutrient signature of dietary generalism in an ecologically diverse primate in the wild. *Behav. Ecol.* 29(4):804–13
120. Takahashi MQ, Rothman JM, Raubenheimer D, Cords M. 2021. Daily protein prioritization and long-term nutrient balancing in a dietary generalist, the blue monkey. *Behav. Ecol.* 32(2):223–35
121. Hou R, Chapman CA, Rothman JM, Zhang H, Huang K, et al. 2021. The geometry of resource constraint: an empirical study of the golden snub-nosed monkey. *J. Anim. Ecol.* 90(3):751–65
122. Eisert R. 2011. Hypercarnivory and the brain: protein requirements of cats reconsidered. *J. Comp. Physiol. B* 181(1):1–17

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

123. Veldhorst MA, Westerterp-Plantenga MS, Westerterp KR. 2009. Gluconeogenesis and energy expenditure after a high-protein, carbohydrate-free diet. *Am. J. Clin. Nutr.* 90(3):519–26
124. Johnson RJ, Stenvinkel P, Martin SL, Jani A, Sánchez-Lozada LG, et al. 2013. Redefining metabolic syndrome as a fat storage condition based on studies of comparative physiology. *Obesity* 21(4):659–64
125. Oda M, Satta Y, Takenaka O, Takahata N. 2002. Loss of urate oxidase activity in hominoids and its evolutionary implications. *Mol. Biol. Evol.* 19(5):640–53
126. Lim JY, Wasserman MD, Veen J, Després-Einspenner M-L, Kissling WD. 2021. Ecological and evolutionary significance of primates' most consumed plant families. *Proc. Biol. Sci.* 288(1953):20210737
127. Sengupta A, Radhakrishna S. 2015. Fruit trait preference in rhesus macaques (*Macaca mulatta*) and its implications for seed dispersal. *Int. J. Primatol.* 36(5):999–1013
128. Cui Z, Shao Q, Grueter CC, Wang Z, Lu J, Raubenheimer D. 2019. Dietary diversity of an ecological and macronutritional generalist primate in a harsh high-latitude habitat, the Taihangshan macaque (*Macaca mulatta tcheliensis*). *Am. J. Primatol.* 81(4):e22965
129. Mc Auley MT. 2022. Dietary restriction and ageing: recent evolutionary perspectives. *Mech. Ageing Dev.* 208:111741
130. Le Couteur DG, Raubenheimer D, Solon-Biet S, de Cabo R, Simpson SJ. 2022. Does diet influence aging? Evidence from animal studies. *J. Intern. Med.* 295(4):400–15
131. Mirzaei H, Raynes R, Longo VD. 2016. The conserved role for protein restriction during aging and disease. *Curr. Opin. Clin. Nutr. Metab. Care* 19(1):74–79
132. Babygirija R, Lamming DW. 2021. The regulation of healthspan and lifespan by dietary amino acids. *Transl. Med. Aging* 5:17–30
133. Wali JA, Ni D, Facey HJW, Dodgson T, Pulpitel TJ, et al. 2023. Determining the metabolic effects of dietary fat, sugars and fat-sugar interaction using nutritional geometry in a dietary challenge study with male mice. *Nat. Commun.* 14:4409
134. Solon-Biet SM, Walters KA, Simanainen UK, McMahon AC, Ruohonen K, et al. 2015. Macronutrient balance, reproductive function, and lifespan in aging mice. *PNAS* 112(11):3481–86
135. Kowald A, Kirkwood TBL. 2015. Evolutionary significance of ageing in the wild. *Exp. Gerontol.* 71:89–94
136. Chiva M. 1997. Cultural aspects of meals and meal frequency. *Br. J. Nutr.* 77(Suppl. 1):S21–28
137. Acosta-Rodríguez V, Rijo-Ferreira F, Izumo M, Xu P, Wight-Carter M, et al. 2022. Circadian alignment of early onset caloric restriction promotes longevity in male C57BL/6J mice. *Science* 376(6598):1192–202
138. Pak HH, Haws SA, Green CL, Koller M, Lavarias MT, et al. 2021. Fasting drives the metabolic, molecular and geroprotective effects of a calorie-restricted diet in mice. *Nat. Metab.* 3(10):1327–41
139. Boyd P, O'Connor SG, Heckman-Stoddard BM, Sauter ER. 2022. Time-restricted feeding studies and possible human benefit. *JNCI Cancer Spectr*. 6(3):pkac032
140. Rynders CA, Thomas EA, Zaman A, Pan Z, Catenacci VA, Melanson EL. 2019. Effectiveness of intermittent fasting and time-restricted feeding compared to continuous energy restriction for weight loss. *Nutrients* 11(10):2442
141. Cui Z, Wang Z, Zhao G, Lu J. 2015. Seasonal changes in activity time allocation in adult female Taihang Mountain macaques: effects of food and temperature. *J. Vet. Sci.* 35(2):138–46
142. Burian RM. 1993. How the choice of experimental organism matters: epistemological reflections on an aspect of biological practice. *J. Hist. Biol.* 26(2):351–67
143. Kellogg EA, Shaffer HB. 1993. Model organisms in evolutionary studies. *Syst. Biol.* 42(4):409–14
144. Bolker JA. 2009. Exemplary and surrogate models: two modes of representation in biology. *Perspect. Biol. Med.* 52(4):485–99
145. Ankeny RA, Leonelli S. 2011. What's so special about model organisms? *Stud. Hist. Philos. Sci. A* 42(2):313–23
146. Bolker JA. 2014. Models in context: biological and epistemological niches. In *Entangled Life*, ed. G Barker, E Desjardins, T Pearce, 4:153–66. Dordrecht: Springer Neth.
147. Wehling M. 2015. *Principles of Translational Science in Medicine: From Bench to Bedside*. Cambridge, MA: Academic. 2nd ed.
148. Green S, Dietrich MR, Leonelli S, Ankeny RA. 2018. 'Extreme' organisms and the problem of generalization: interpreting the Krogh principle. *Hist. Philos. Life Sci.* 40(4):65

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17

149. Nawata CM, Pannabecker TL. 2018. Mammalian urine concentration: a review of renal medullary architecture and membrane transporters. *J. Comp. Physiol. B* 188(6):899–918
150. Duffy MA, García-Robledo C, Gordon SP, Grant NA, Green DA, et al. 2021. Model systems in ecology, evolution, and behavior: a call for diversity in our model systems and discipline. *Am. Nat.* 198(1):53–68
151. Gosby AK, Conigrave AD, Raubenheimer D, Simpson SJ. 2014. Protein leverage and energy intake. *Obes. Rev.* 15(3):183–91
152. Raubenheimer D, Gosby AK, Simpson SJ. 2015. Integrating nutrients, foods, diets, and appetites with obesity and cardiometabolic health. *Obesity* 23(9):1741–42

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:11:17